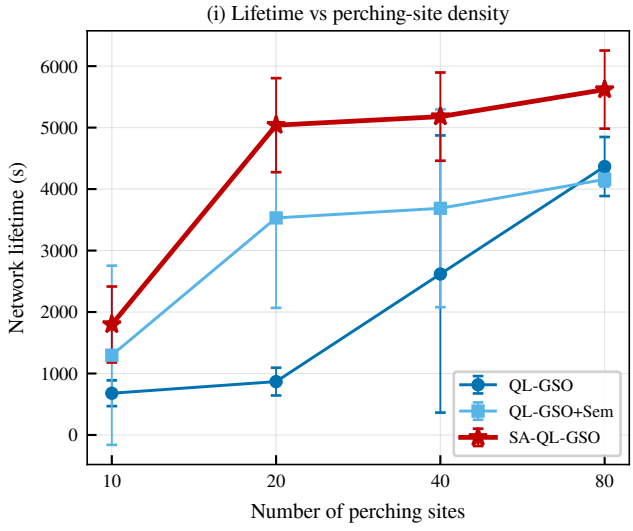
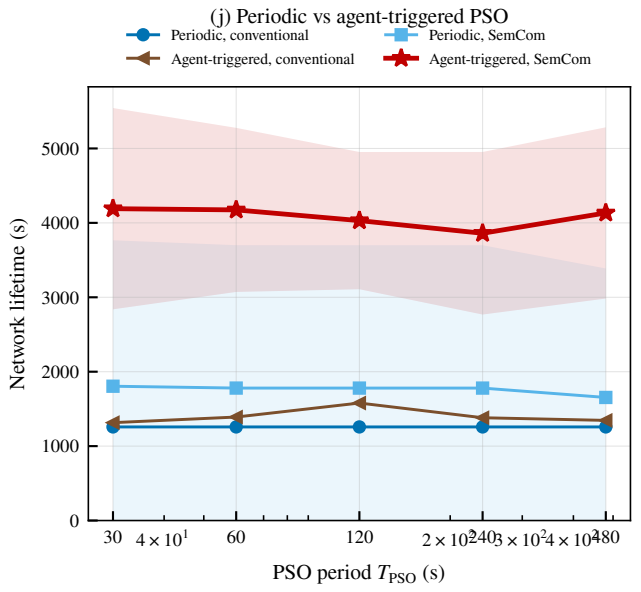
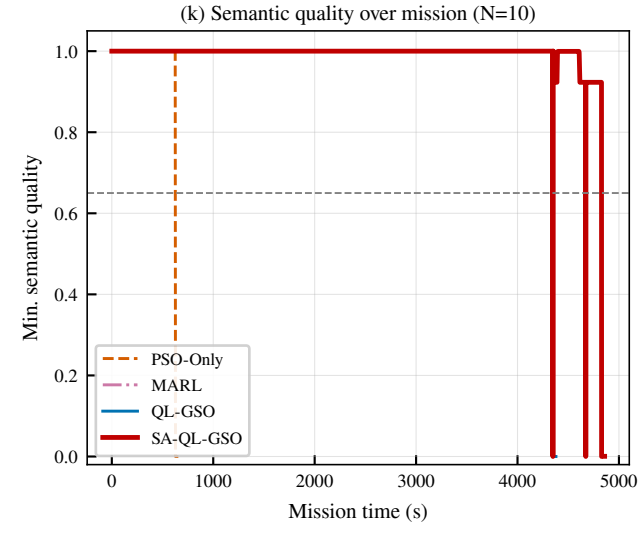
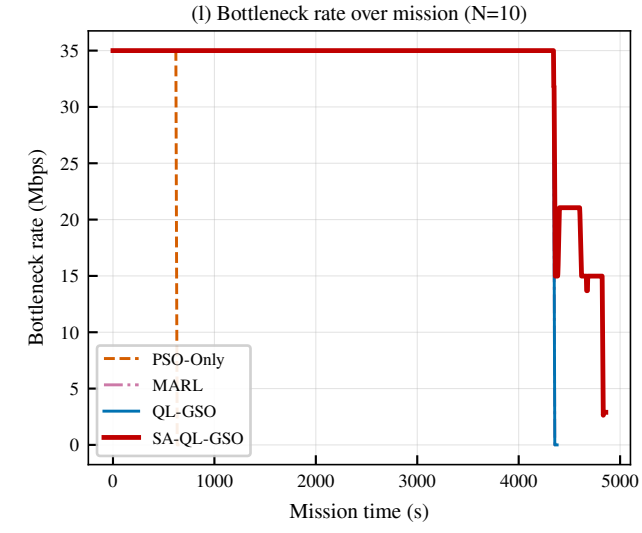
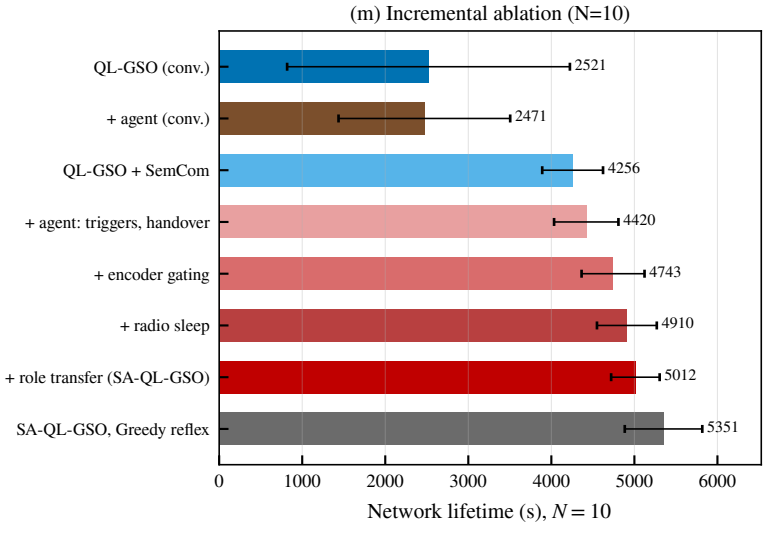
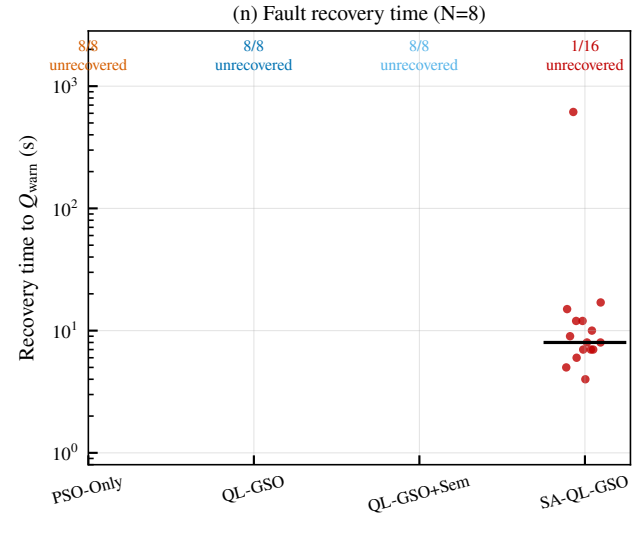

(i) Lifetime vs perching-site density
Network lifetime (s)
Number of perching sites
QL-GSO
QL-GSO+Sem
SA-QL-GSO

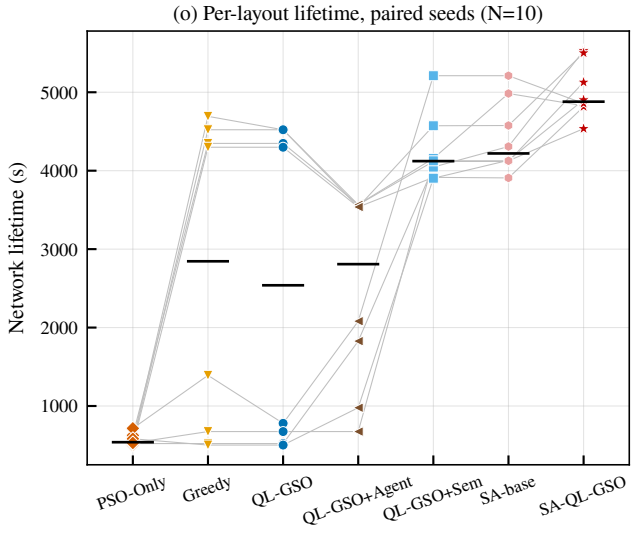

(j) Periodic vs agent-triggered PSO
Periodic, conventional
Periodic, SemCom
Agent-triggered, conventional
Agent-triggered, SemCom
Network lifetime (s)
PSO period T_{PSO} (s)


(k) Semantic quality over mission (N=10)
Min. semantic quality
Mission time (s)
PSO-Only
MARL
QL-GSO
SA-QL-GSO


(l) Bottleneck rate over mission (N=10)
Bottleneck rate (Mbps)
Mission time (s)
PSO-Only
MARL
QL-GSO
SA-QL-GSO


(m) Incremental ablation (N=10)
QL-GSO (conv.)
+ agent (conv.)
QL-GSO + SemCom
+ agent: triggers, handover
+ encoder gating
+ radio sleep
+ role transfer (SA-QL-GSO)
SA-QL-GSO, Greedy reflex
2521
2471
4256
4420
4743
4910
5012
5351
Network lifetime (s), $N = 10$


(n) Fault recovery time (N=8)
8/8 unrecovered
8/8 unrecovered
8/8 unrecovered
1/16 unrecovered
Recovery time to Q_{warn} (s)
PSO-Only
QL-GSO
QL-GSO+Sem
SA-QL-GSO


(o) Per-layout lifetime, paired seeds (N=10)
Network lifetime (s)
PSO-Only
Greedy
QL-GSO
QL-GSO+Agent
QL-GSO+Sem
SA-base
SA-QL-GSO